



Databases on Azure – what's new? (2025 Edition)

Haiko Hertes, Azure Architect & Principal Consultant, Microsoft Azure MVP + MCT
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- Seit 2019 bei SoftwareOne
- Principal Consultant & Architect
- Azure Consulting Team
- Microsoft MVP, YouTuber, Blogger, Conference Speaker
- Familienvater, Offizier d.R., Holzwurm



www.hertes.net



about.me/haiko.hertes



twitter.com/HHertes



youtube.com/c/HaikoHertes

Haiko Hertes

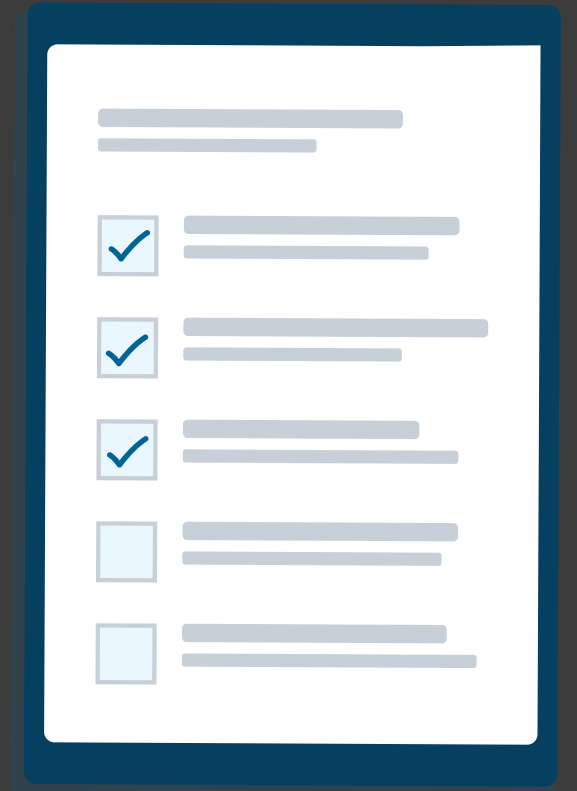
Cloud Architect / Principal Consultant



AGENDA

- 01** Azure Database for Postgres
- 02** Azure Database for MySQL
- 03** Azure SQL Database / Managed Instance
- 04** Azure SQL Server VMs
- 05** Miscellaneous

Some of the features are in Preview – check before using them!



01



Azure Database for Postgres

Azure Database for PostgreSQL

- Single Server deployment option retired March 2025!
- Latest supported minor versions as of today:
 - 17.6, 16.10, 15.14, 14.19, 13.22 – all older versions are out of support!
 - [Unterstützte Versionen von PostgreSQL - Azure Database for PostgreSQL](#)
- **Reservations for v5 Flexible Server**
 - [Prepay for Azure Database for PostgreSQL flexible server compute resources with reserved capacity](#)
- Online Migration from Single Server to Flexible Server
 - [Migrate To Azure Database For Postgresql Flexible Server From Single Server - Migrate from Single Server](#)

Azure Database for PostgreSQL

- Support to modify several performance parameters
 - [Server parameters in Azure Database for PostgreSQL flexible server](#)
- **On-demand backups (physical snapshots)**
 - [Backup and restore - Azure Database for PostgreSQL](#)
- Elastic clusters for horizontal scaling
 - [Postgres horizontal scaling with elastic clusters on Azure Database for PostgreSQL Flexible server](#)
- Foreign Data Wrapper (FDW) extension for Oracle
 - [Allow extensions](#)

Azure Database for PostgreSQL

- Index tuning (analyze queries, recommend indexes to create or to drop)
 - [Index tuning - Azure Database for PostgreSQL](#)
- **Discovery of PostgreSQL on Arc-enabled servers**
 - [PostgreSQL and MySQL installation identification by Azure Arc-enabled servers](#)
- **Enhanced performance guidance in Azure Advisor**
 - [Boost Your Postgres Server on Azure with Enhanced Azure Advisor Performance Recommendations!](#)

Azure Database for PostgreSQL

Advisor recommendations

Azure Advisor provides personalized recommendations to reduce costs, increase security, optimize performance and reliability and achieve operational excellence.

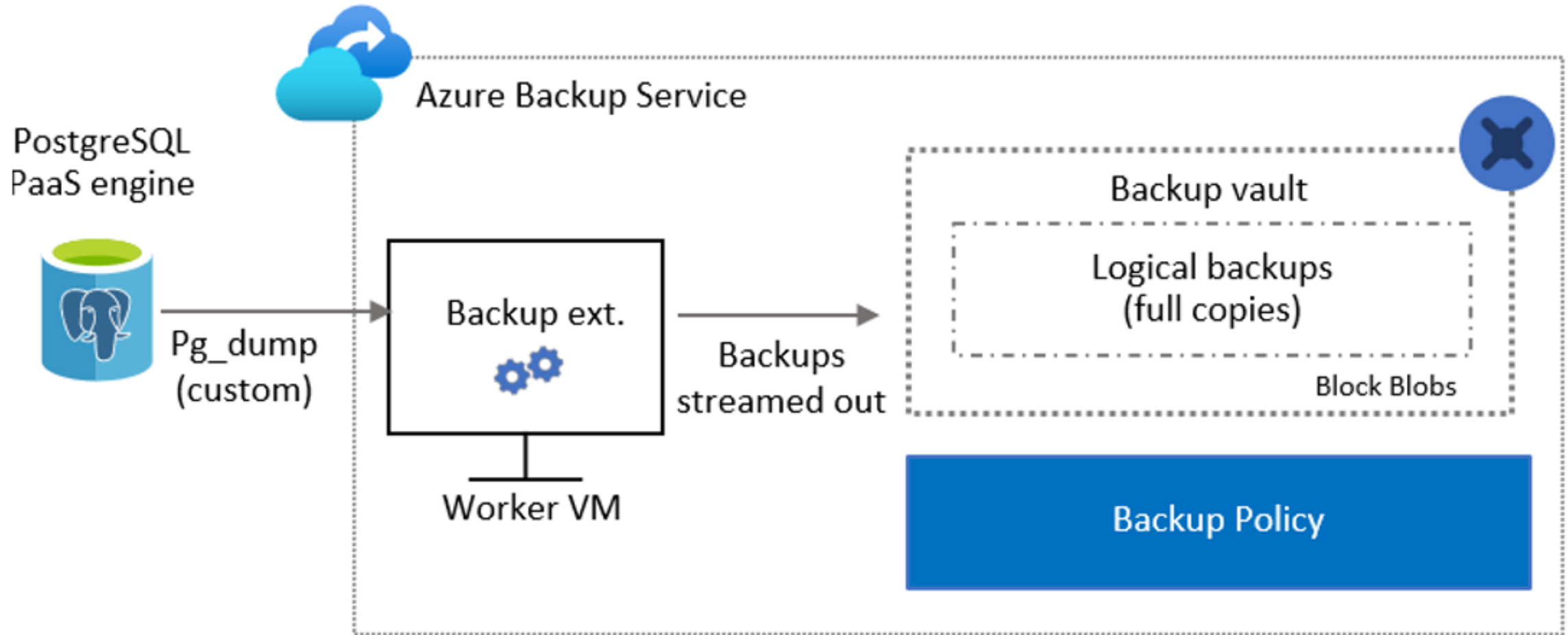
All	Cost	Security	Reliability	Performance	Operational Excellence
 12	 -	 -	 1	 11	 -

Impact ↑↓	Recommendation ↑↓	Category ↑↓	Impacted resources ↑↓
High	Review your server for too frequent checkpoints	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for log_min_duration_statement turned ON	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for inactive Logical Replication Slots	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for high CPU utilization	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for bloat ratio greater than 80%	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for autovacuum turned OFF	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for long running transactions	Performance	1 Azure Database for PostgreSQL flexible server
High	Review your server for log_statement_stats turned ON	Performance	1 Azure Database for PostgreSQL flexible server
Medium	Configure geo redundant backup storage	Reliability	1 Azure Database for PostgreSQL flexible server
Medium	Review your server for log_error_verbosity set to VERBOSE	Performance	1 Azure Database for PostgreSQL flexible server

Azure Database for PostgreSQL

- **Vaulted Backups using Azure Backup and Backup Vaults**
- Long-term backup retention using Azure Backup
 - [About Azure Database for PostgreSQL Flexible server backup - Azure Backup](#) (both features above)
 - [About Azure Database for PostgreSQL Flexible server backup - Azure Backup](#)
- In-place major version upgrade from at least v14 to v17
 - [Upgrade to PostgreSQL 17 on Azure Flexible Server](#)

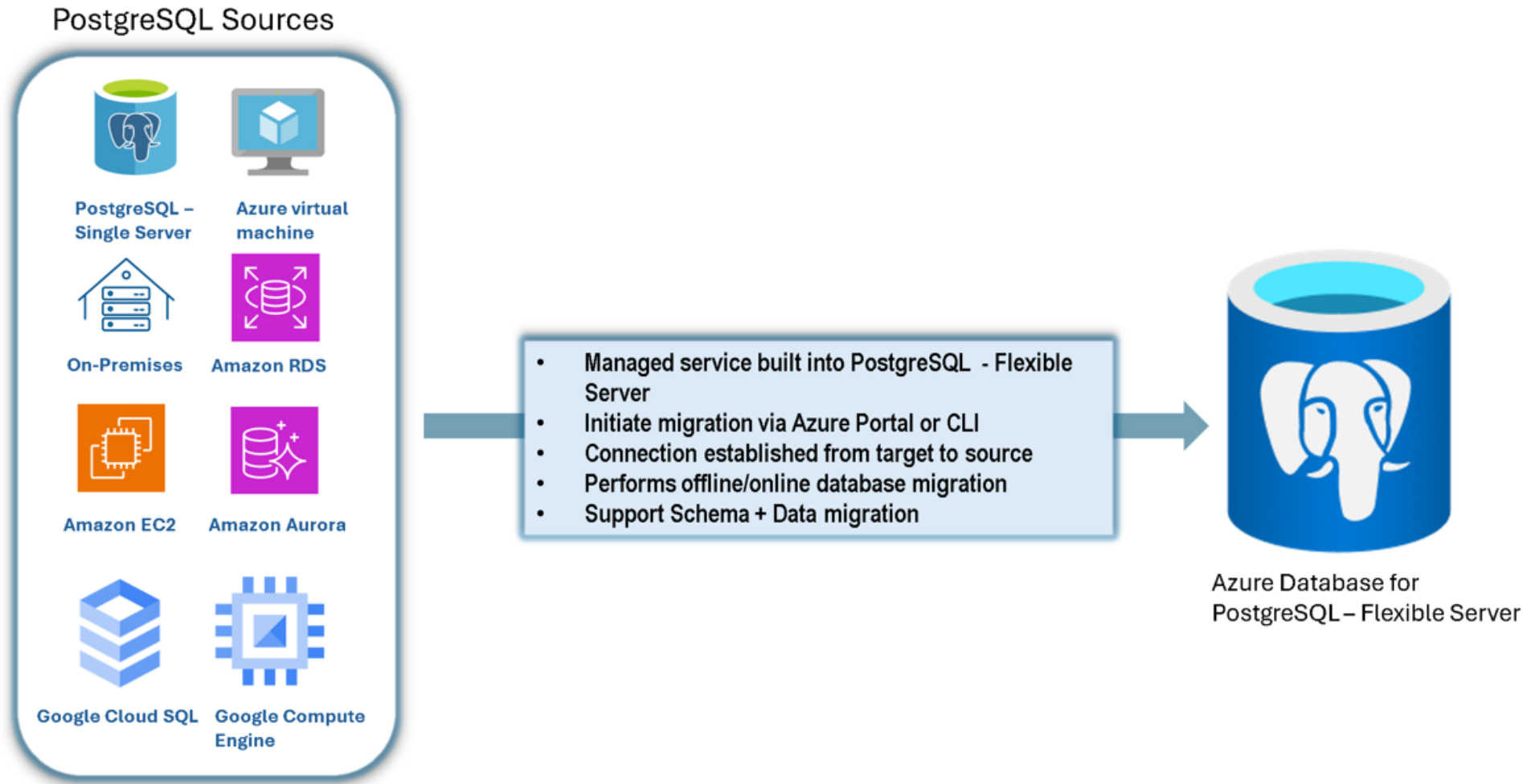
Azure Database for PostgreSQL



Azure Database for PostgreSQL

- Managed Identity (user-assigned and system-assigned)
 - [How to Connect from Azure Data Factory to Azure Database for PostgreSQL - Azure Database for PostgreSQL](#)
- **Confident computing with hardware-based trusted execution environment (TEE)**
 - [Azure Confidential Computing - Azure Database for PostgreSQL](#)
- Migration Service for PostgreSQL 17
 - [Migration Service in Azure Database for PostgreSQL - Azure Database for PostgreSQL](#)

Azure Database for PostgreSQL



A woman with dark hair in a ponytail, wearing a light pink button-down shirt, is looking down at a tablet computer she is holding. She is standing in a server room with rows of server racks visible in the background. The racks have blue and green indicator lights. The overall lighting is dim, with the server lights providing the main illumination.

02

Azure Database for MySQL

Azure Database for MySQL

- Latest supported minor versions as of today:
 - 5.7, 8.0, 8.4 in GA, 9.3 in Preview – all older versions are out of support!
 - [Version Support Policy](#)
 - Extended Support starting in spring 2026 for up to 3 additional years (Will have a price tag!)
- **Zone-resilient Business Critical tier by default at no additional costs** (affects storage in first place)
 - [Service Tiers - Azure Database for MySQL | Microsoft Learn](#)
- Accelerated Logs and support for Customer Managed Keys
 - [Accelerated Logs Feature in Azure Database for MySQL - Flexible Server](#)

Azure Database for MySQL

- **User-managed plugin enablement without server restart**
 - [Built-in Store Procedure in Azure Database for MySQL - Flexible Server - Azure Database for MySQL](#)
- Triggers and Bindings for Azure Functions
 - [Azure Database for MySQL Bindings for Functions](#)
- Unified Migration Experience in Azure DBMS
 - [Migrate from MySQL to Azure Database for MySQL Online Using DMS Physical Migration via the Azure Portal](#)
- **MySQL Discovery and Assessment in Azure Migrate**
 - [MySQL Discovery and Assessment in Azure Migrate - Public Preview](#)

Azure Database for MySQL

MySQL Assessment ...

mysql-assessment

 Settings  Recalculate assessment  Delete |  Export  Refresh

Last calculated on 4/3/2025 6:41:54 pm

Overview Instances to Azure Database for MySQL

Assessed workloads

Servers	MySQL instances	User databases	MySQL instances past EOL	Discovery success
7	8	70	4	100%

Migration scenarios ⓘ

▼ All instances to Azure Databases for MySQL

Readiness



[View Details](#)

Sources ready to migrate ⓘ

Ready  0 Ready with conditions  1

Monthly cost estimate (USD)

For 8 instances



Azure Database for MySQL

- **Configure backup interval for automated backups**
 - [Backup and Restore - Azure Database for MySQL](#)
- Trigger and delete on-demand backups
 - [Backup and Restore - Azure Database for MySQL](#)
- Near-Zero Downtime for (planned) maintenance for HA-enabled instances
 - [Scheduled Maintenance - Azure Database for MySQL](#)
- **Self Heal to resolve common issues without support cases**
 - [Server self-heal for Azure Database for MySQL - Flexible Server - Azure Database for MySQL](#)

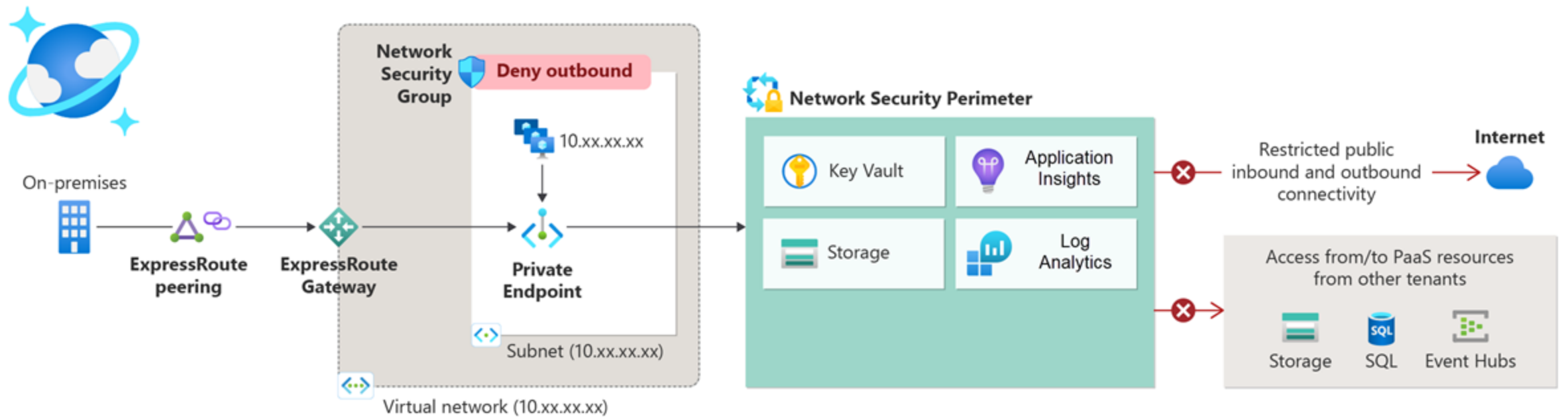
03

Azure SQL Database / Managed Instance

Azure SQL Database

- Azure SQL Database Hyperscale with some performance / sizing improvements (DB Size, Log Rate)
 - [Announcing enhancements to Azure SQL Database Hyperscale](#)
- **One and three year reservations for Azure SQL Database now available, also for zone-redundant GP tier**
 - [Introducing Reservations for Zone Redundant General Purpose SQL Database](#)
- **Network Security Perimeter for Azure SQL Database**
 - [What is a network security perimeter? - Azure Private Link](#)

Azure SQL Database



Azure SQL Database

- **Filtering egress traffic to Storage Accounts so safeguard against data exfiltration**
 - [Service endpoint policies for Azure Storage now generally available in SQL Managed Instance subnets](#)
 - [Configure service endpoint policies - Azure SQL Managed Instance](#)
- Now 10 free Azure SQL Databases per subscription, each with 32GB GP serverless with 100.000 vCore-seconds per month
 - [Deploy for Free - Azure SQL Database | Microsoft Learn](#)
- Convert Azure SQL Database to Hyperscale with explicit cutover control
 - [Improving the conversion to Hyperscale with greater efficiency | Microsoft Community Hub](#)
- ...

Azure SQL Database

- **Lower & simplified pricing for Azure SQL DB and Managed Instance Backup Storage**
 - [Simplified & lower pricing for Azure SQL Database and Azure SQL Managed Instance backup storage](#)
- Retirement of Fsv2-series Hardware on 1st Oct 2026 – no new deployments after 1st Oct 2025
 - [vCore purchasing model - Azure SQL Database](#) / [Azure updates](#)
- RA-GZRS as backup storage type for non-hyperscale tiers
- **Azure SQL Hub experience in Azure Portal**
 - [Introducing the Azure SQL hub: A simpler, guided entry into Azure SQL | Microsoft Community Hub](#)

Azure SQL - All

Database options to optimize performance, capacity, and cost

Experience the performance, scale, and security of the Microsoft Azure cloud with fully managed Azure SQL database services. [Learn more](#)



Which option is best for you?

Answer a few questions about your database needs, and we'll find the right option for you.

[Find the right option](#)



Compare options side by side

Compare features, specs, and key differences — or just start creating.

[Compare options](#)



Azure SQL Database Hyperscale

Platform as a service

Build AI-powered applications on fully managed, high-performance databases with extreme scalability.

- Supports up to 128 TB of high performance storage that automatically grows as database size increases
- Handles unpredictable and growing workloads
- Faster transaction commit times and higher log throughput

[Show options](#)



Azure SQL Managed Instance

Platform as a service

Migrate on-premise instances and databases to a fully managed SQL environment with minimal changes to your applications.

- Near 100% compatibility with SQL Server Enterprise Edition
- Automated backups, patching, monitoring, and maintenance
- Integrates with Azure services like Defender for SQL and Azure Monitor for enhanced protection and insights

[Show options](#)

[Try for free](#)



Azure SQL Database

Platform as a service

Build applications using a fully managed SQL database with built-in performance and reliability.

- Automated backups, patching, high availability, and disaster recovery
- Database-level control and elastic scaling—efficiently match performance and cost to workload needs
- Integrates with Azure services like Microsoft Defender for SQL and Azure Monitor for enhanced protection and insights

[Show options](#)

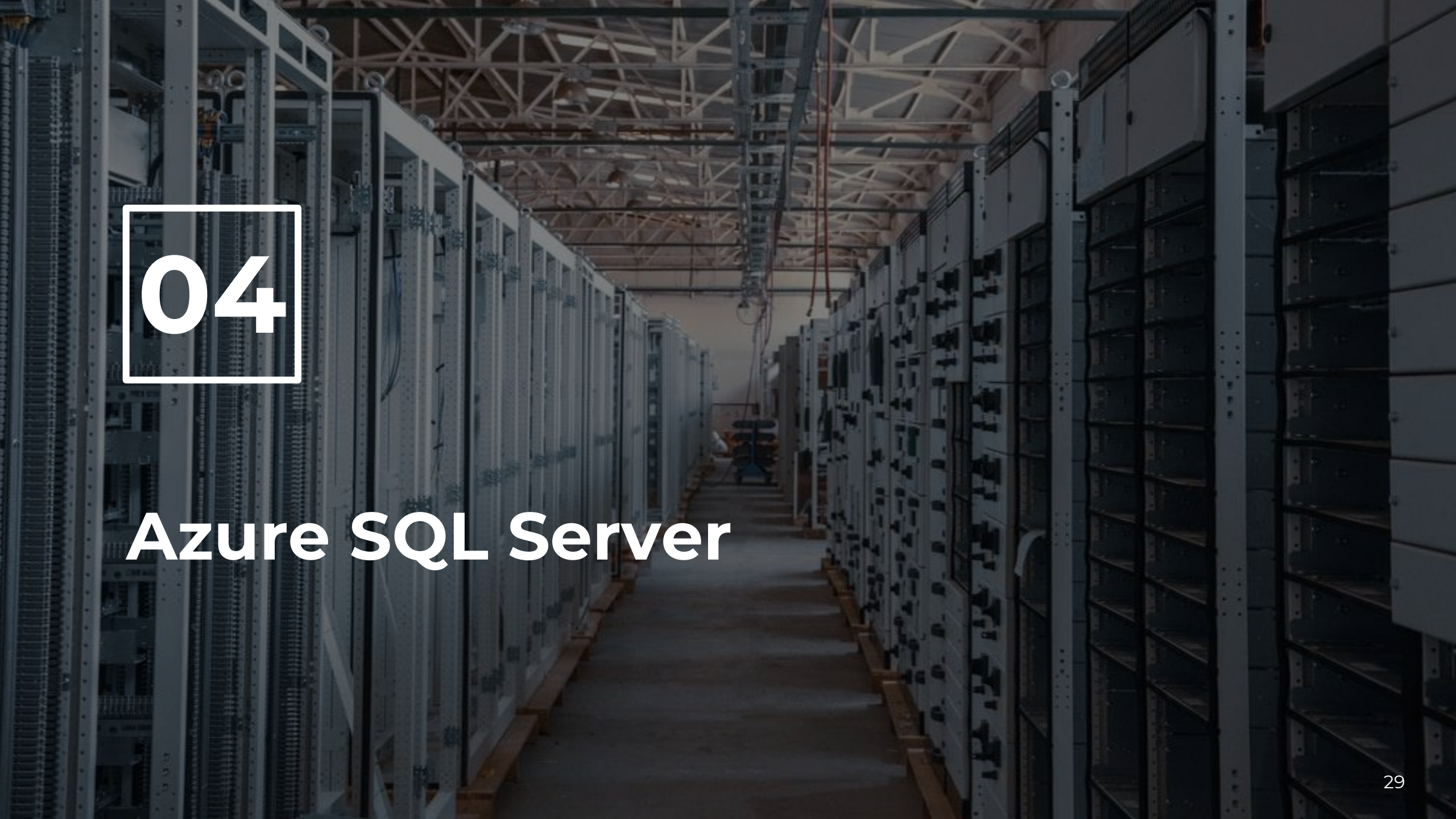
[Try for free](#)

Azure SQL Managed Instance

- **Instance Pools are GA** (allowing to share resources across multiple instances, allowing 2-vCore-instances, faster deployment for additional instances and cost-savings!)
 - [Azure SQL Managed Instance pools: General Availability](#)
 - [What is an instance pool? - Azure SQL Managed Instance](#)
- Use of native Windows principals now available
 - [Native Windows principals for Azure SQL Managed Instance are now generally available](#)
- Azure SQL Managed Instance Free Offer (One GP MI for 12 month per Subscription)
 - [Free SQL Managed Instance offer is now generally available](#)

Azure SQL Managed Instance

- Adjust RAM per vCore in Azure SQL Managed Instance for better cost-efficiency.
- [Unlocking More Power with Flexible Memory in Azure SQL Managed Instance](#)



04

Azure SQL Server

Azure SQL Server

- Modernization Advisor for SQL Server on Azure VMs
 - [Modernization Advisor \(Preview\) - SQL Server on Azure VMs](#)
- I/O Performance Analysis for SQL Server on Azure Virtual Machines
 - [Analyze I/O Performance - SQL Server on Azure VMs](#)

Comparison results

Summary of benefits running SQL Server workloads on Azure SQL Managed Instance compared to this SQL Virtual Machine:

Feature description	 SQL Server on Azure VM	 Azure SQL Managed Instance
Resource name	contoso-sql-vm	-
SQL Server version	SQL Server 2019	SQL Server 2022 compatible, up-to-date
SQL Server compatibility level	100, 110, 120, 130	100, 110, 120, 130, 140, 150, 160
SQL Server Edition	Enterprise	Enterprise
Hardware profile	4 vCores, 16 MB RAM (Standard D4s v3)	Gen5, 4 vCores
Storage size	2048 GB	2048 GB
License type ⓘ	Pay as you go	Pay as you go
High availability SLA starting with 99.99% ⓘ	-	✓

Ready to be modernized!

Take advantage of integrated benefits of running your SQL Server workloads on a fully managed [Azure SQL Managed Instance](#) platform as a service (PaaS)

- ✓ 24x7 service health monitoring and healing
- ✓ Automated OS and SQL Server patching and upgrades
- ✓ Automated backups with short and long-term retention
- ✓ Built-in multi-layered enterprise security
- ✓ Risk free evaluation for up to 12 months



Microsoft Azure

Home > SQL virtual machine

contoso-sql
SQL virtual machine

Search

Settings

Configure

Storage

Updates

Backups

High Availability

SQL best practices
assessment

Modernization Advisor
(preview)

Properties

Security

Azure Metric	Metric description	Problematic condition	I/O throttling conclusions
Disk latency (preview)	The average time to complete IOs for the data disk during the monitored period. Values are in milliseconds.	> 500 milliseconds in a consecutive 5-minute period	There is a latency problem for the system to further investigate potential throttling.
VM Cached IOPS Consumed Percentage	The percentage calculated by the total IOPS completed over the max cached virtual machine IOPS limit.	>= 95% in a consecutive 5-minute period	There is VM throttling. The application running on the SQL virtual machine is fully utilizing the maximum cached IOPS capacity available to the virtual machine - the storage demands of the application exceed the cached IOPS provided by the virtual machine's underlying storage configuration.
VM Cached Bandwidth Consumed Percentage	The percentage calculated by the total disk throughput completed over the max cached virtual machine throughput.	>= 95% in a consecutive 5-minute period	There is VM throttling. The application running on the SQL virtual machine is utilizing the maximum available cached disk bandwidth for data transfer - the data transfer demands of the application exceed the cached bandwidth resources provided by the virtual machine's underlying storage configuration.
VM Uncached IOPS Consumed Percentage	The percentage calculated by the total IOPS on a virtual machine completed over the max uncached virtual machine IOPS limit.	>= 95% in a consecutive 5-minute period	There is VM throttling. The application running on the SQL virtual Machine is utilizing the maximum allowable uncached IOPS capacity available to the virtual machine- the storage demands of the application exceed the uncached IOPS resources provided by the virtual machine's underlying storage configuration.
VM Uncached Bandwidth Consumed Percentage	The percentage calculated by the total disk throughput on a virtual machine completed over the max provisioned virtual machine throughput.	>= 95% in a consecutive 5-minute period	There is VM throttling. The application running on the SQL virtual machine is utilizing the maximum allowable uncached disk bandwidth for data transfer – the data transfer demands of the application exceed the uncached bandwidth resources provided by the virtual machine's underlying storage configuration.
Data Disk IOPS Consumed Percentage	The percentage calculated by the data disk IOPS completed over the provisioned data disk IOPS.	>= 95% in a consecutive 5-minute period	There is data disk throttling. The application running on the SQL virtual machine is hitting the IOPS limit for the provisioned data disk - the storage demands of the application exceed the performance capabilities of the chosen disk configuration.
Data Disk Bandwidth Consumed Percentage	The percentage calculated by the data disk throughput completed over the provisioned data disk throughput.	>= 95% in a consecutive 5-minute period	There is data disk throttling. The application running on the SQL virtual machine is hitting the IOPS limit for the provisioned data disk - the storage demands of the application exceed the performance capabilities of the chosen disk configuration.

05

Miscellaneous

```
128 font-family:
129
130 }
131
132 em.mail{
133     background: url(../img/mailico.png) no-repeat center;
134     display: inline-block;
135     width: 12px;
136     height: 14px;
137     float: left;
138     margin: 2px 7px 0 0;
139 }
140 em.phone{
141     background: url(../img/phoneico.png) no-repeat center;
142     display: inline-block;
143     width: 20px;
144     height: 18px;
145     float: left;
146     margin: 2px 7px 0 0;
147 }
```

Miscellaneous

- Azure Data Studio (ADS) will retire Feb 2026 – transition to VS Code
 - [What's Happening to Azure Data Studio? - Azure Data Studio](#)
- New Azure Region in Austria
 - [Microsoft eröffnet Rechenzentrumsregion in Österreich - Source EMEA](#)

Thank You!

Questions?

one
software

Vielen Dank!
Ich freue mich auf Feedback!

Scan
mich!





Haiko Hertes

Haiko.Hertes@SoftwareONE.com

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